1.Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)

- To find the probability of rolling a 3 exactly 5 times in 30 rolls, use Excel's `BINOM.DIST` function.

Step 1: Simulate 30 dice rolls

In cell A1, enter:

```
=RANDBETWEEN(1,6)
```

Drag the formula down to A30 to generate 30 random dice rolls.

Step 2: Count how many times 3 appears

Use the formula:

```
=COUNTIF(A1:A30,3)
```

This counts the number of times the number 3 appears in your 30 rolls.

Step 3: Calculate the probability using BINOM.DIST

Enter this formula in Excel:

```
=BINOM.DIST(5,30,1/6,FALSE)
```

Here:

- 5 = number of successes (rolling a 3)
- 30 = total number of trials (rolls)
- 1/6 = probability of rolling a 3 on one roll
- FALSE = probability of exactly 5 successes

Step 4: Final answer

The probability is approximately:

$$P(X = 5) = \binom{30}{5}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^{25}$$

\[\boxed{P(X=5)\approx 0.1921} \]

Answer: 19.21%

2.Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

- Generate 100 random values using RAND() in Excel

Step 1: Generate 100 values

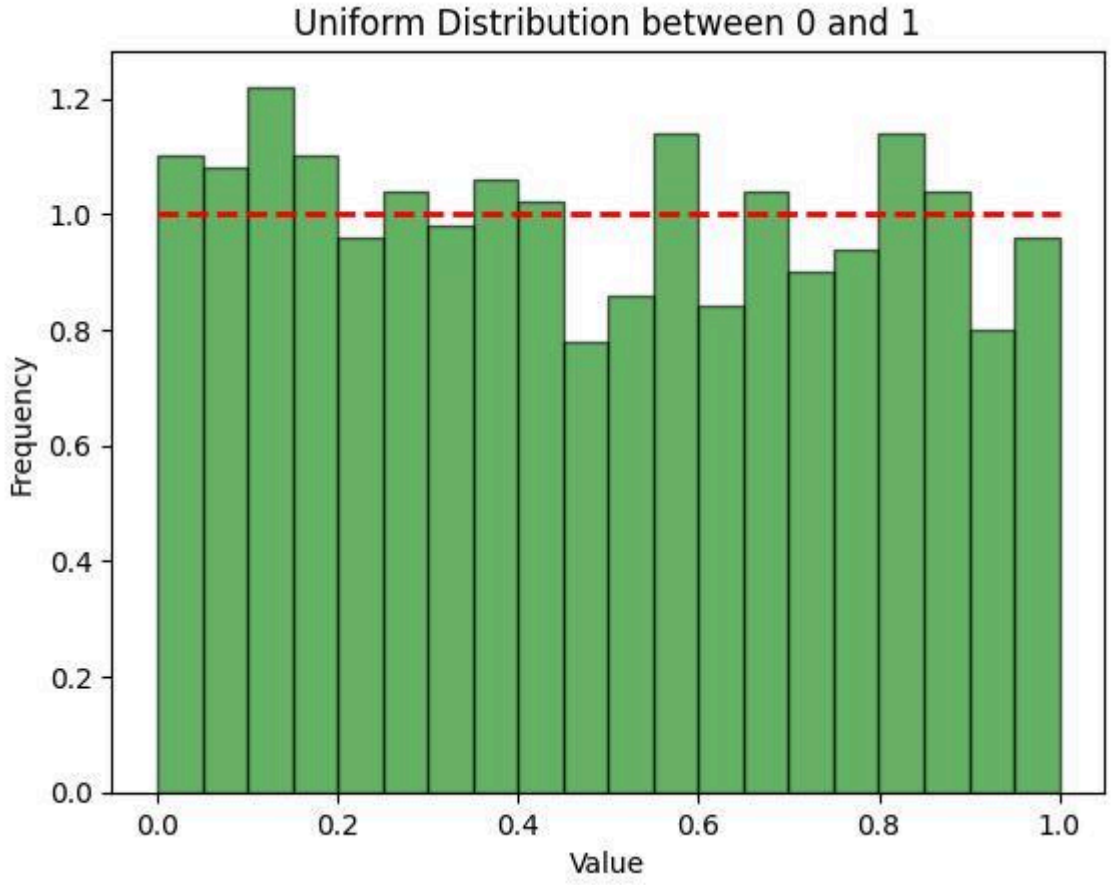
1. Open Excel and type `Random Value` in cell A1.
2. In cell A2, enter the formula:

```
=RAND()
```

3. Drag the formula down to cell A101 to generate 100 random values between 0 and 1.

Note: `RAND()` generates a new random value whenever Excel recalculates.

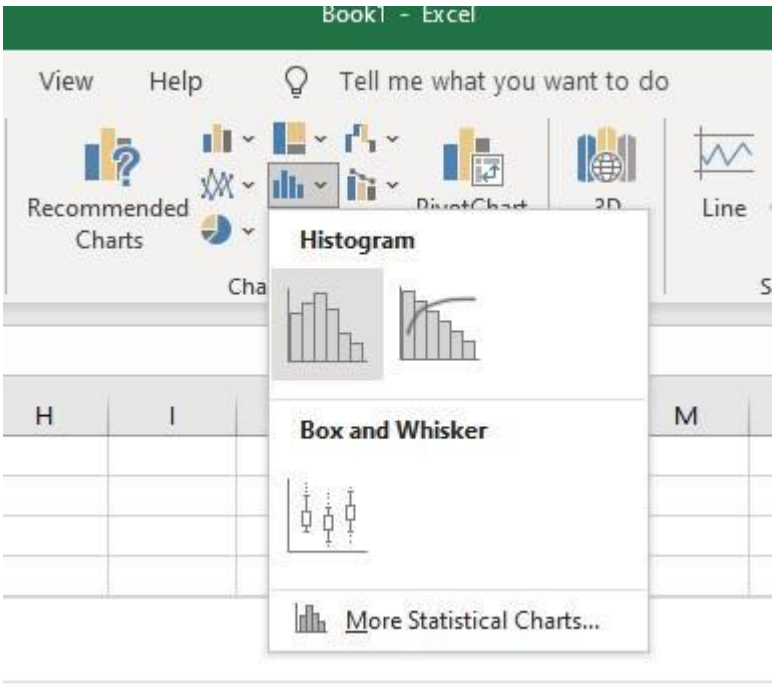
Step 2: Plot a histogram



1. Select the data

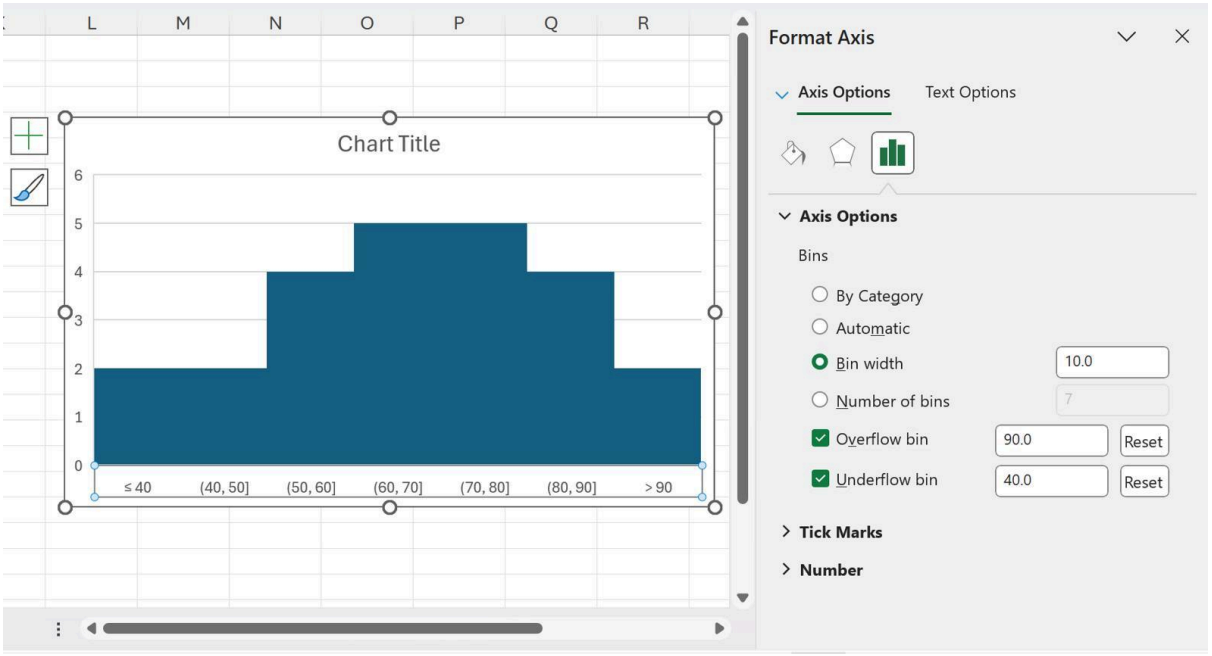
Select cells A2:A101.

DISTRIBUTION



2. Insert a histogram

Go to Insert → Insert Statistic Chart → Histogram.



3. Format the chart

Set the bin width to `0.1` to create 10 bins from 0 to 1.

Step 3: Describe the shape of the distribution

Expected histogram: Uniform distribution

DISTRIBUTION

Illustrative histogram — actual bar heights will vary.

Simulated RAND() Values

Illustrative counts for 100 random values grouped into 10 equal-width bins.

04812160.0–0.10.1–0.20.2–0.30.3–0.40.4–0.50.5–0.60.6–0.70.7–0.80.8–0.90.9–1.0

3.A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

- The dataset has:
- Mean (μ) = 50
- Standard deviation (σ) = 5
- Range = 45 to 55

Step 1: Calculate the Z-scores

Use the formula:

$$Z = \frac{X-\mu}{\sigma}$$

For $X = 45$:

$$Z = \frac{45-50}{5} = -1$$

For $X = 55$:

$$Z = \frac{55-50}{5} = 1$$

Step 2: Apply the Empirical Rule

In a normal distribution, approximately 68.27% of values lie within one standard deviation of the mean.

Since 45 and 55 are one standard deviation below and above the mean, respectively:

$$P(45 \leq X \leq 55) = P(-1 \leq Z \leq 1)$$

\[\boxed{68.27\%} \]

4.What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

- 1. What is Standardization (Z-score)?

Standardization is a statistical technique used to transform data so that its mean becomes 0 and its standard deviation becomes 1.

A Z-score tells us how many standard deviations a data value lies above or below the mean.

- A positive Z-score means the value is above the mean.
- A negative Z-score means the value is below the mean.
- A Z-score of 0 means the value is equal to the mean.

2. Formula for Z-score

\[\boxed{Z=\frac{X-\mu}{\sigma}} \]

Where:

- Z = standardized value (Z-score)
- X = original data value
- μ = mean of the dataset
- σ = standard deviation of the dataset

$z = \frac{x-\mu}{\sigma}$
$z = \frac{(1)-(0)}{1} = 1$
\(\Phi(z)\approx \text{84.1\%}\)
x
μ
σ
μxΦ(z)
Give feedback

3. Example of Standardization

Suppose a student's marks have a mean of 50 and a standard deviation of 10. The student scores 70 marks.

Given:

- $X = 70$
- $\mu = 50$
- $\sigma = 10$

Calculation:

$$Z = \frac{70-50}{10} = \frac{20}{10} = 2$$

Interpretation: The student's score is 2 standard deviations above the mean.

Similarly, if a student scores 40:

$$Z = \frac{40-50}{10} = -1$$

This means the score is 1 standard deviation below the mean.

4. How does standardization transform a dataset?

Consider the original dataset:

$$10,\ 20,\ 30,\ 40,\ 50$$

The mean is 30. Using the population standard deviation, $\sigma \approx 14.14$, we calculate each Z-score.

Original value (X)	**Z-score**
10	-1.41
20	-0.71
30	0.00
40	0.71
50	1.41

After standardization:

- The mean becomes approximately 0.
- The standard deviation becomes 1.
- The relative positions of the values remain unchanged.
- The data is expressed on a common scale.

Standardization does not necessarily make the dataset follow a normal distribution; it only changes its scale and center.

5. Why is standardization important in data analysis?

1. Comparing different datasets: It allows us to compare values measured on different scales, such as exam marks and salaries.
2. Identifying outliers: Extremely high or low Z-scores can help identify unusual observations.
3. Machine learning: Methods such as K-nearest neighbors (KNN), support vector machines (SVM), and principal component analysis (PCA) can benefit from standardized features.
4. Statistical analysis: It helps determine how far an observation is from the mean.
5. Fair comparison: It helps compare relative performance when datasets have different means and standard deviations.

6. Standardization in Excel

Suppose the original values are in cells A2:A6.

- Calculate the mean: `=AVERAGE(A2:A6)`
- Calculate the population standard deviation: `=STDEV.P(A2:A6)`
- Calculate the Z-score for A2:

`=(A2-AVERAGE(`A`2:`A`6))/STDEV.P(`A`2:`A`6)`

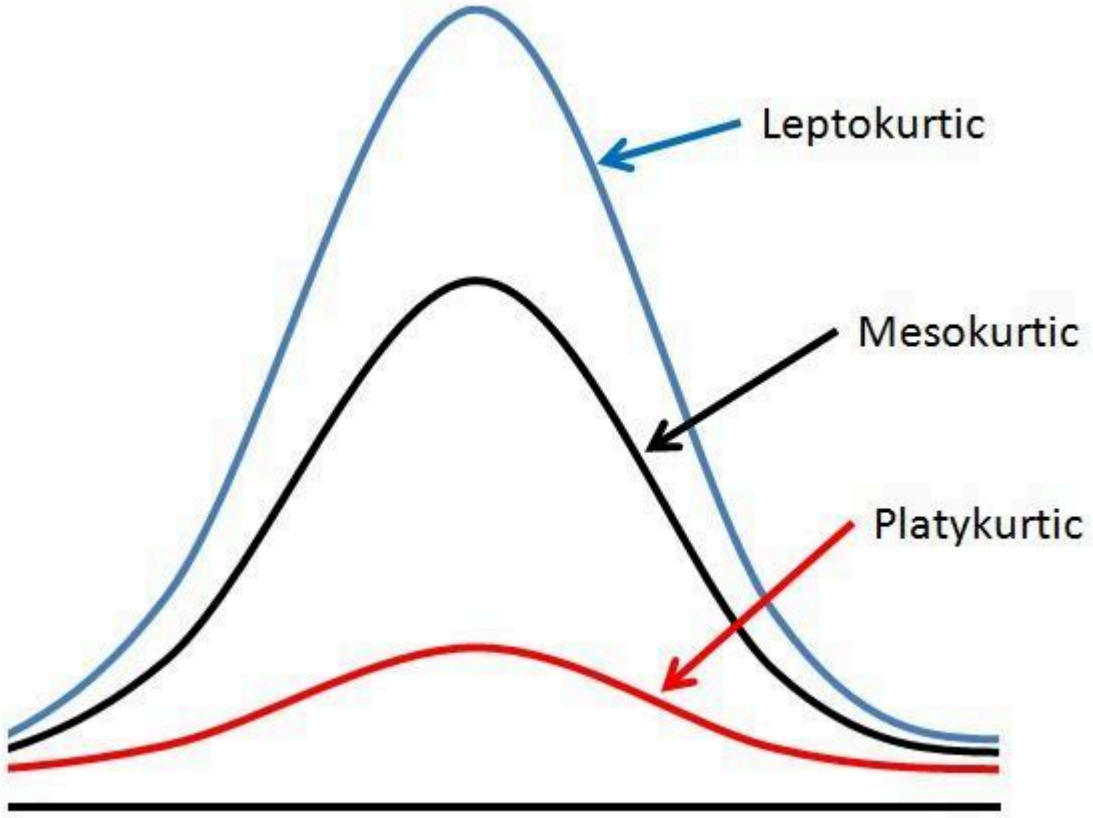
Drag the last formula down to standardize all values.

5.What is Kurtosis and their type?

- Kurtosis is a statistical measure that describes the tailedness of a data distribution. It helps us understand whether a dataset has heavier or lighter tails and how likely it is to contain extreme values (outliers).

There are three main types of kurtosis:

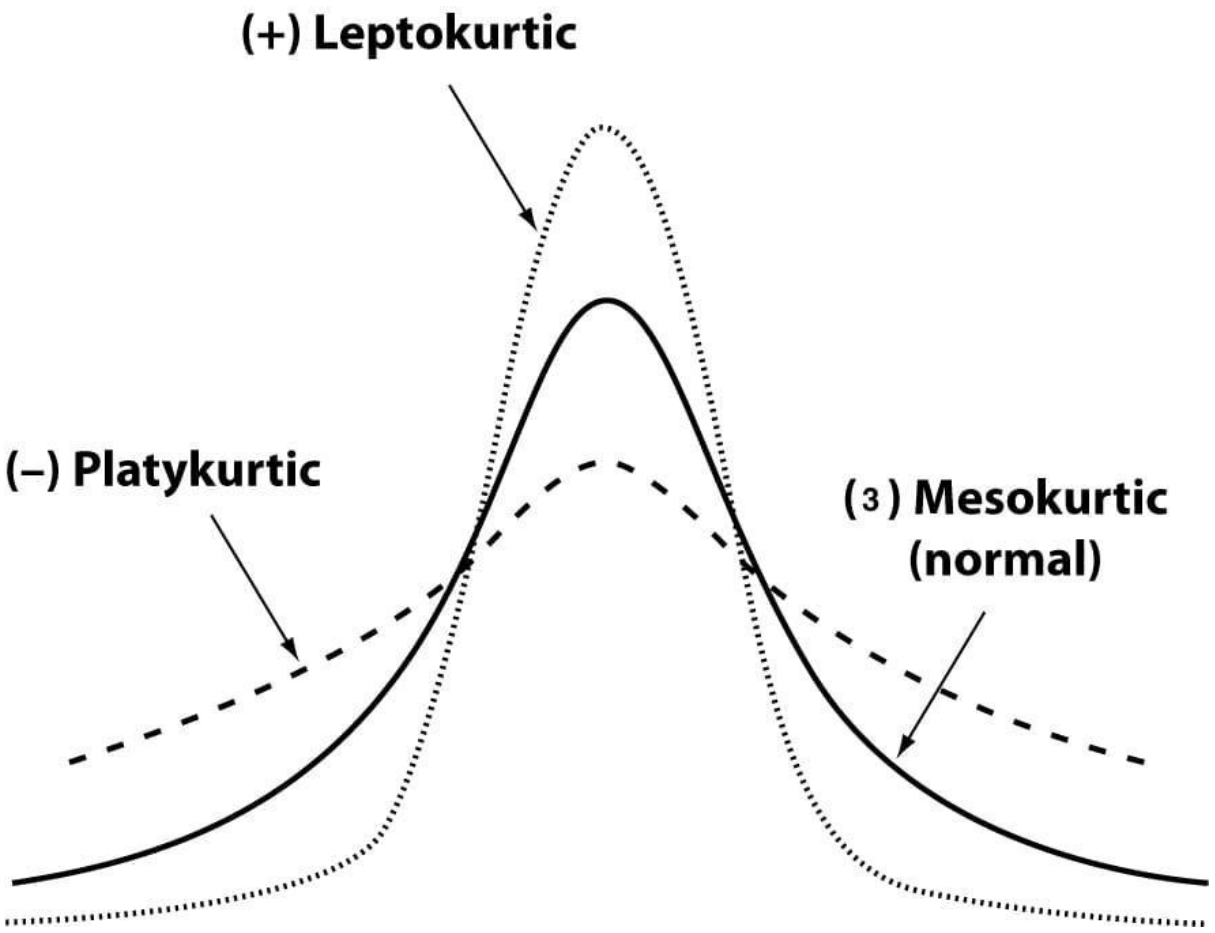
DISTRIBUTION



1. Leptokurtic

- Has heavier tails than a normal distribution.
- Indicates a greater tendency for extreme values.
- Excess kurtosis: greater than 0.

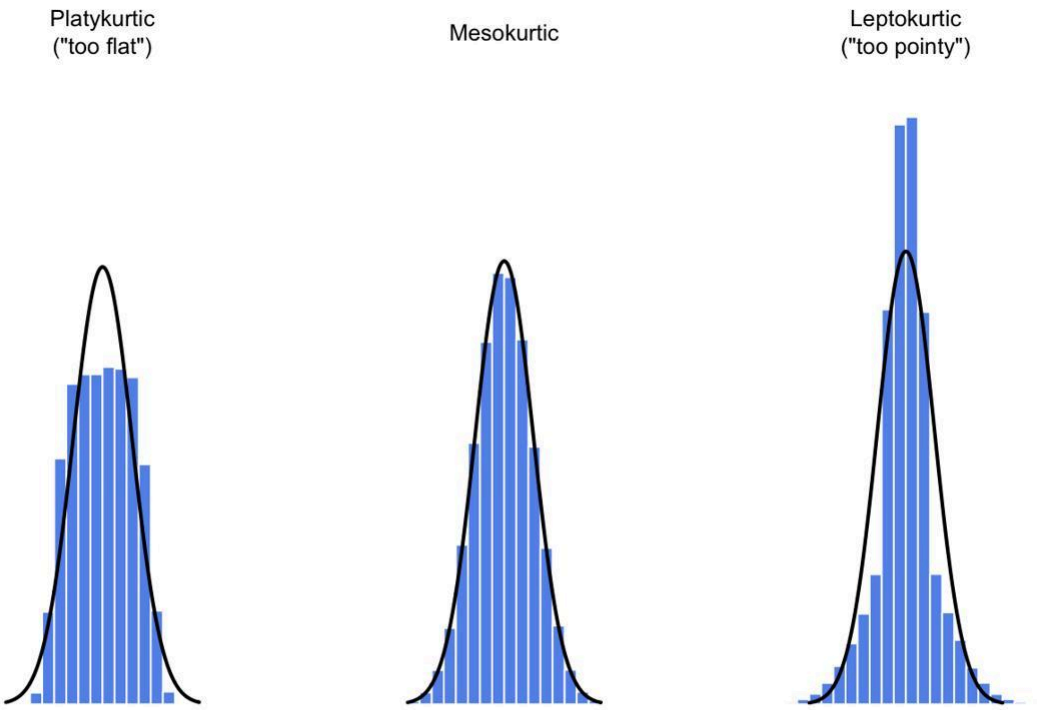
DISTRIBUTION



2. Mesokurtic

- Has the same kurtosis as a normal distribution.
- Serves as the standard comparison.
- Excess kurtosis: equal to 0.

DISTRIBUTION



3. Platykurtic

- Has lighter tails than a normal distribution.
- Often has a flatter central peak.
- Excess kurtosis: less than 0.

Formula for Kurtosis

Population kurtosis is calculated using the fourth central moment:

$$Kurtosis = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Where:

- X = data value
- μ = mean
- σ = standard deviation
- E = expected value

For a normal distribution, kurtosis is 3. Excess kurtosis is calculated as:

$$ExcessKurtosis = Kurtosis - 3$$

Therefore, a normal distribution has excess kurtosis of 0.

Kurtosis in Excel

If your data is in cells A1:A100, use:

```
=KURT(A1:A100)
```

Excel's `KURT` function returns the sample excess kurtosis.

6.Explain why the uniform distribution is a good model for the outcome of rolling a fair die.

- A uniform distribution is a probability distribution in which all possible outcomes have an equal probability of occurring.

A fair six-sided die is a good example of a discrete uniform distribution because each face has an equal chance of appearing when the die is rolled.

1. Possible outcomes

When a fair die is rolled, the possible outcomes are:

$$S = \{1, 2, 3, 4, 5, 6\}$$

There are six possible outcomes, and each has an equal probability.

2. Probability of each outcome

The probability formula is:

$$P(X = x) = \frac{1}{Totalnumberofoutcomes}$$

Therefore,

$$P(X = x) = \frac{1}{6} \approx 0.1667$$

Each number has a probability of approximately 16.67%.

DISTRIBUTION

Outcome	Probability
1	1/6 (16.67%)
2	1/6 (16.67%)
3	1/6 (16.67%)
4	1/6 (16.67%)
5	1/6 (16.67%)
6	1/6 (16.67%)
Total	1 (100%)

3. Why is it a uniform distribution?

- Equal probability: Every face has the same chance of appearing.
- Finite outcomes: There are six distinct possible outcomes.
- Constant probability: No number is more likely than another.
- Total probability: The probabilities of all outcomes add up to 1.

7.Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

- To calculate the probability of getting at least 8 successes in 15 trials with a success probability of 0.5, use Excel's `BINOM.DIST` function.

Step 1: Identify the values

- Number of trials (n) = 15
- Probability of success (p) = 0.5
- Minimum number of successes (x) = 8

We need to find:

$$P(X \geq 8)$$

Step 2: Enter the Excel formula

Because `BINOM.DIST` with `TRUE` calculates the cumulative probability, we subtract the probability of 7 or fewer successes from 1.

Excel formula:

```
=1-BINOM.DIST(7,15,0.5,TRUE)
```

Step 3: Result

$$P(X \geq 8) = 1 - P(X \leq 7)$$

\[\boxed{P(X\geq8)\approx0.5000} \]

Final answer: 50%

Explanation: Since the success probability is 0.5 and there are 15 trials, the binomial distribution is symmetric around 7.5. Therefore, the probability of getting at least 8 successes is exactly 50%.

8.How does log transformation help in stabilizing variance and making data more normally distributed?

- Log transformation is a technique in data analysis in which each data value is replaced by its logarithm. It is commonly used to reduce skewness, stabilize variance, and make data more closely resemble a normal distribution.

1. Formula for Log Transformation

$$Y = \log(X)$$

Where:

- X = original data value
- Y = transformed data value

In Excel, you can use:

- `=LN(A2)` for the natural logarithm.
- `=LOG10(A2)` for the base-10 logarithm.

The original values must be positive for these formulas to work directly.

2. How does it stabilize variance?

In many datasets, larger values have greater variability than smaller values. This is called heteroscedasticity.

Log transformation compresses large values more than small values, helping reduce the difference in spread across the data.

For example:

Original value	LN-transformed value
1	0.00
10	2.30
100	4.61
1,000	6.91
10,000	9.21

Notice that the original values increase dramatically, but their logarithms increase much more gradually.

3. How does it make data more normally distributed?

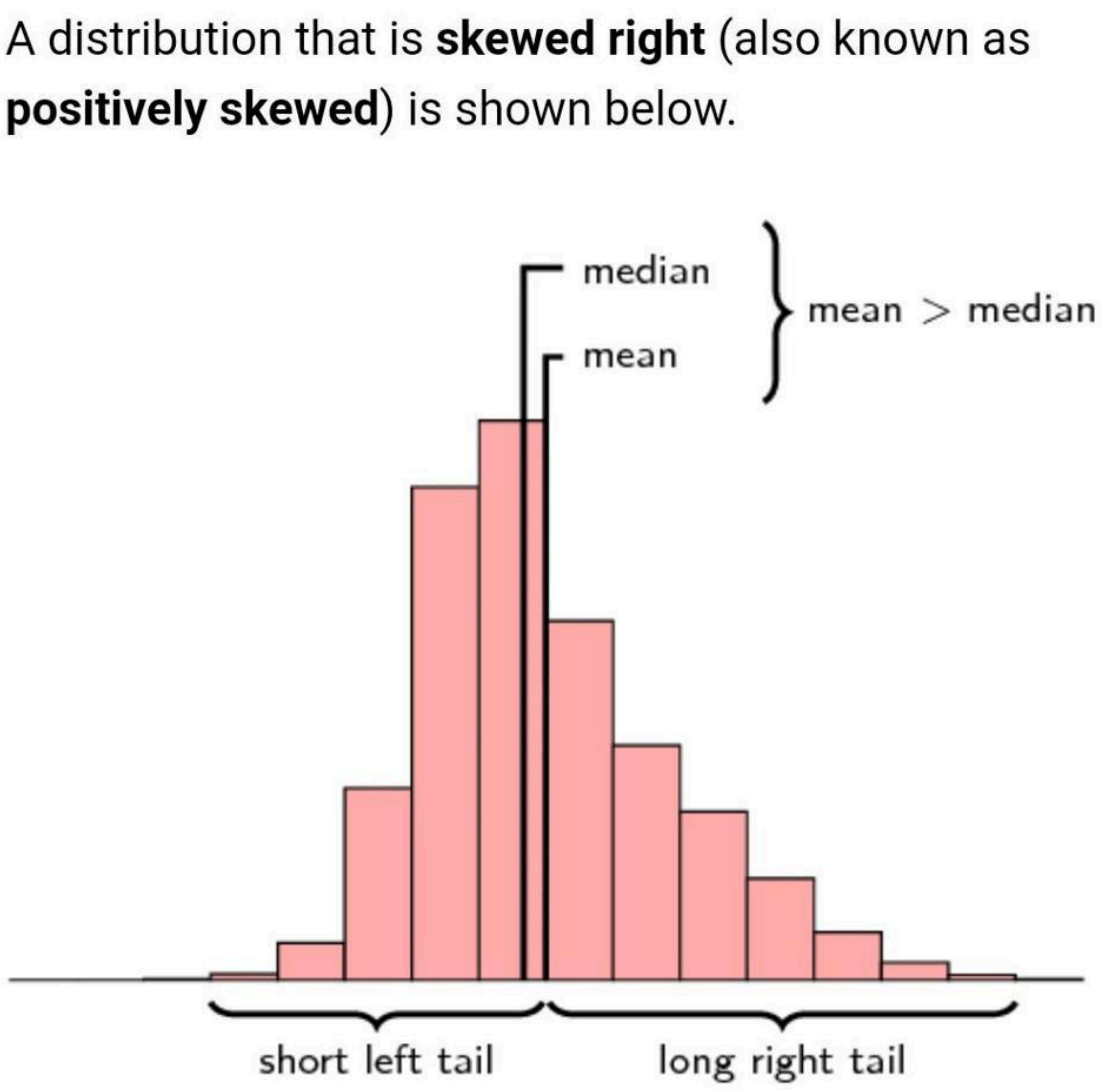
DISTRIBUTION

Log transformation is particularly useful for positively skewed data, where a few very large values create a long right tail.

It can help by:

- Compressing extreme high values.
- Reducing right skewness.
- Making the distribution more symmetric.
- Improving the suitability of data for statistical methods that assume approximately normal errors.

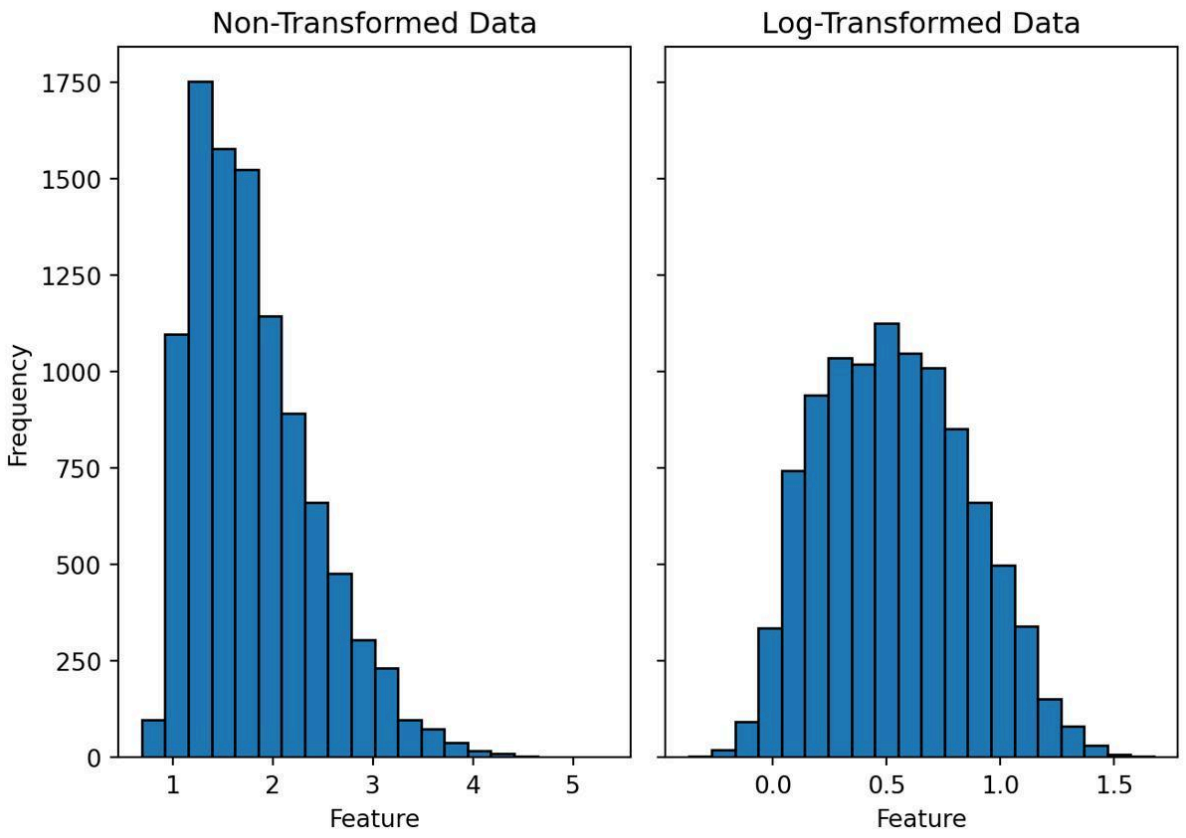
For example, income, property prices, and company sales often have right-skewed distributions and may become more symmetric after a log transformation.



Before transformation

The data has a long right tail because a few observations are much larger than most values.

DISTRIBUTION



After transformation

The distribution may become more symmetric, with extreme values having less influence on its shape.

Note: Log transformation does not guarantee normality or constant variance. Its effectiveness depends on the data.

4. Practical example in Excel

Suppose sales values are in cells A2:A6:

Sales	Log-transformed sales
100	4.61
200	5.30

500	6.21
1,000	6.91
10,000	9.21

In cell B2, enter:

```
=LN(A2)
```

Drag the formula down to transform all sales values. You can then compare histograms of the original and transformed data.

5. Why is log transformation important?

1. Variance stabilization: It can reduce variability that increases with the mean.
2. Skewness reduction: It often reduces positive skewness.
3. Outlier influence: It compresses extreme large values.
4. Statistical modeling: It can help meet assumptions of certain regression and statistical methods.
5. Interpretation: Differences on the log scale can represent relative or percentage changes on the original scale.